

Equivalence Classes and the Quotient Set $A/\sim$

CS1231S AY26/27 Semester 1

1 Recap: equivalence relations

A relation $\sim$ on a set A is an **equivalence relation** if, for all $a, b, c \in A$:

- **Reflexive:** $a \sim a$.
- **Symmetric:** if $a \sim b$ then $b \sim a$.
- **Transitive:** if $a \sim b$ and $b \sim c$ then $a \sim c$.

Think of $a \sim b$ as saying “ a and b are the same (a general notion of equal) in some chosen respect” (same remainder, same sign, ...).

2 Equivalence classes

Definition. Let $\sim$ be an equivalence relation on A and let $a \in A$. The **equivalence class** of a is the set of everything related to a :

$$[a] = \{x \in A : x \sim a\}.$$

Three facts to remember:

1. $a \in [a]$, so no class is empty (by reflexivity).
2. $[a] = [b]$ exactly when $a \sim b$.
3. Two classes are either *identical* or *disjoint*. They never partly overlap.

Fact 2 means a class can have many names: any of its elements can be used as the **representative** written inside the brackets.

3 The quotient set $A/\sim$

Definition. The **quotient set** of A by $\sim$ is the set of all equivalence classes:

$$A/\sim = \{[a] : a \in A\}.$$

So $A/\sim$ is a *set of sets*. Its elements are whole classes (a set of elements that are all equivalent under $\sim$), not elements of A .

Definition. A **partition** of A is a collection of non-empty subsets of A such that every element of A lies in *exactly one* of them. Equivalently, the subsets are pairwise disjoint and their union is A .

Theorem. If $\sim$ is an equivalence relation on A , then $A/\sim$ is a partition of A .

Proof. We need to show two things:

- For any $a \in A$, there exists some set $X \in A/\sim$ such that $a \in X$

- For any $X, Y \in A/\sim$, $X \cap Y = \emptyset$ or $X = Y$

To prove 1, by definition of $A/\sim$, notice that for any $a \in A$, $[a] \in A/\sim$. Hence, 1 has been proven. To prove 2, it follows from facts 2 and 3 above.

4 Example 1: congruence mod 5

Let $A = \mathbb{N} = \{0, 1, 2, 3, \dots\}$ and define

$$a \sim b \iff 5 \mid (a-b) \iff a \text{ and } b \text{ leave the same remainder on division by 5.} \iff a \equiv b \pmod{5}$$

Proof sketch of it being an equivalence relation:

- Reflexive: $a - a = 0$ and $5 \mid 0$.
- Symmetric: if $5 \mid (a - b)$ then $5 \mid (b - a)$, since $b - a = -(a - b)$.
- Transitive: if $5 \mid (a - b)$ and $5 \mid (b - c)$ then $5 \mid (a - c)$, since $a - c = (a - b) + (b - c)$.

The classes. There is one class for each possible remainder:

$$\begin{aligned} [0] &= \{0, 5, 10, 15, \dots\} \\ [1] &= \{1, 6, 11, 16, \dots\} \\ [2] &= \{2, 7, 12, 17, \dots\} \\ [3] &= \{3, 8, 13, 18, \dots\} \\ [4] &= \{4, 9, 14, 19, \dots\} \end{aligned}$$

The quotient set.

$$\mathbb{N}/\sim = \{[0], [1], [2], [3], [4]\}.$$

The set $\mathbb{N}$ is infinite, but $\mathbb{N}/\sim$ has only 5 elements.

It is a partition.

- Each class is non-empty.
- The classes are pairwise disjoint: a number cannot have two different remainders.
- Their union is $\mathbb{N}$: every number has *some* remainder in $\{0, 1, 2, 3, 4\}$.

For instance, $17 = 5 \cdot 3 + 2$, so 17 lies in $[2]$ and in no other class.

Remark. Since $7 \sim 2$, we have $[7] = [2] = [12]$. These are three names for the same class, with three different representatives.

5 Example 2: same sign

Let $A = \mathbb{R} \setminus \{0\}$ and define

$$a \sim b \iff ab \geq 0.$$

Since neither a nor b is 0, this really says $ab > 0$, i.e. a and b have the same sign.

It is an equivalence relation.

- Reflexive: $a \cdot a = a^2 > 0$.
- Symmetric: $ab = ba$.
- Transitive: if $ab > 0$ and $bc > 0$ then $(ab)(bc) = ac \cdot b^2 > 0$, and $b^2 > 0$, so $ac > 0$.

The classes. There are exactly two:

$$[1] = \{x \in \mathbb{R} : x > 0\}, \quad [-1] = \{x \in \mathbb{R} : x < 0\}.$$

The quotient set.

$$A/\sim = \{[1], [-1]\} = \{(0, \infty), (-\infty, 0)\}.$$

Every non-zero real is positive or negative, never both, so these two classes partition A .

Remark (Why remove 0?). On all of $\mathbb{R}$ the relation $ab \geq 0$ is *not* transitive: $1 \sim 0$ and $0 \sim -1$, but $1 \not\sim -1$. With 0 included, we get no partition.

6 Example 3: differing by an even integer

Let $A = \mathbb{R}$ and define

$$a \sim b \iff a - b = 2k \text{ for some } k \in \mathbb{Z}.$$

In words: a and b are related when you can get from one to the other in steps of 2.

It is an equivalence relation.

- Reflexive: $a - a = 0 = 2 \cdot 0$.
- Symmetric: if $a - b = 2k$ then $b - a = 2(-k)$, and $-k \in \mathbb{Z}$.
- Transitive: if $a - b = 2k$ and $b - c = 2m$ then $a - c = 2(k + m)$, and $k + m \in \mathbb{Z}$.

The classes. The class of a is everything reachable from a in steps of 2:

$$[a] = \{a + 2k : k \in \mathbb{Z}\} = \{\dots, a - 4, a - 2, a, a + 2, a + 4, \dots\}.$$

For instance:

$$\begin{aligned} [0] &= \{\dots, -4, -2, 0, 2, 4, \dots\} && \text{(the even integers)} \\ [1] &= \{\dots, -3, -1, 1, 3, 5, \dots\} && \text{(the odd integers)} \\ [0.5] &= \{\dots, -3.5, -1.5, 0.5, 2.5, 4.5, \dots\} \end{aligned}$$

The quotient set. Every real number x can be written in exactly one way as $x = r + 2k$ with $k \in \mathbb{Z}$ and $0 \leq r < 2$. So each class has exactly one representative in the interval $[0, 2)$, and

$$\mathbb{R}/\sim = \{[r] : 0 \leq r < 2\}.$$

Unlike Examples 1 and 2, there are infinitely many classes: one for each $r \in [0, 2)$.

It is a partition. Every real lies in exactly one class, the one named by its r :

- $7.3 = 1.3 + 2 \cdot 3$, so $7.3 \in [1.3]$.
- $-0.5 = 1.5 + 2 \cdot (-1)$, so $-0.5 \in [1.5]$.

7 Operations on classes and well-definedness

Arithmetic with classes mod 5. Take Example 1, where $\mathbb{N}/\sim = \{[0], [1], [2], [3], [4]\}$. We would like to add and multiply the classes themselves, using the rules

$$[a] + [b] = [a + b], \quad [a] \cdot [b] = [ab].$$

For instance, $[2] + [4] = [6] = [1]$ and $[2] \cdot [4] = [8] = [3]$.

Why there is something to check. These rules are stated using representatives, but a class has many names: $[2] = [7]$ and $[4] = [9]$. The rule really says “pick a representative of each class, add them, and take the class of the result”. If different choices of representative could lead to different classes, the rule would not give a single answer, so it would not define an operation on classes at all.

An operation on classes is **well defined** if the answer depends only on the classes and not on the representatives chosen. For the rules above, this means:

$$\text{if } a \sim a' \text{ and } b \sim b', \text{ then } a + b \sim a' + b' \text{ and } ab \sim a'b'.$$

Mod 5: both operations are well defined. First a quick test with the other names for $[2]$ and $[4]$:

$$[7] + [9] = [16] = [1], \quad [7] \cdot [9] = [63] = [3],$$

the same answers as before. In general, suppose $a \sim a'$ and $b \sim b'$, so $a - a' = 5k$ and $b - b' = 5m$ for some $k, m \in \mathbb{Z}$. Then:

- Addition: $(a + b) - (a' + b') = 5k + 5m = 5(k + m)$, so $a + b \sim a' + b'$.
- Multiplication: $ab - a'b' = (a - a')b + a'(b - b') = 5(kb + a'm)$, and $kb + a'm \in \mathbb{Z}$, so $ab \sim a'b'$.

So $[a] + [b]$ and $[a] \cdot [b]$ give the same class whichever representatives we use.

Even differences: addition works, multiplication fails. Now take Example 3, where $a \sim b$ on $\mathbb{R}$ means $a - b = 2k$ for some $k \in \mathbb{Z}$, and try the same two rules.

- Addition is well defined, by the same argument: if $a - a' = 2k$ and $b - b' = 2m$ then $(a + b) - (a' + b') = 2(k + m)$.
- Multiplication is *not* well defined. We have $[0.5] = [2.5]$, since $2.5 - 0.5 = 2$. But

$$[0.5] \cdot [0.5] = [0.25] \quad \text{while} \quad [2.5] \cdot [0.5] = [1.25],$$

and $[0.25] \neq [1.25]$ because $1.25 - 0.25 = 1$ is not of the form $2k$. The same two classes give two different “products”, so the rule $[a] \cdot [b] = [ab]$ does not define an operation on $\mathbb{R}/\sim$.

Takeaway. Whenever a rule on $A/\sim$ is written in terms of representatives, check that changing the representatives does not change the answer.

8 Summary

- $[a]$ is the set of all elements related to a . a is a representative of that set (the equivalence class). There can be multiple representatives.
- $A/\sim$ is the set of all classes. Its elements are subsets of A .
- $[a] = [b] \iff a \sim b$.
- The classes partition A : every element lies in exactly one class.
- Always check reflexive, symmetric and transitive first. If any one fails, there are no equivalence classes and no quotient set.
- When an operation (this denotes any operator) $\oplus$ is defined on representatives of equivalence classes, like $[a] \oplus [b] = [a \oplus b]$, you have to check if $a \sim a'$ and $b \sim b'$, then $a \oplus b \sim a' \oplus b'$